

PhishAware: Phishing Awareness Simulation & Incident Response

A Controlled Educational Simulation and SOC Incident Management Platform

Candidate: Cybersecurity Intern	Track: Capstone Task 5
Date: September 2026	Classification: Educational / Authorized
Framework: Flask, SQLite, Bootstrap 5	Status: Completed & Verified

Executive Summary

Social engineering and email phishing remain the leading attack vector exploited in enterprise breaches. The PhishAware platform provides an end-to-end, controlled educational environment that models the full cybersecurity lifecycle: planning, scenario creation, simulated dispatch, real-time telemetry, automated anomaly detection, containment, eradication, recovery, and post-incident analysis. Operating strictly in a safe localhost sandbox with synthetic users, the application strictly guarantees zero credential harvesting or password collection. In the primary baseline campaign, 10 synthetic targets yielded an 80% open rate, a 40% click rate, and a 60% reporting rate, triggering automated Incident INC-001 which was contained within 2 minutes 15 seconds.

1. Objectives & Safety Architecture

The project was engineered to satisfy all internship capstone criteria under strict ethical constraints:

- **Zero Credential Collection:** No password input fields or credential storage mechanisms exist.
- **Synthetic Test Targets:** 10 fictional user accounts across multiple corporate departments.
- **Safe Local Routing:** All verification links safely route to educational awareness landing pages.
- **End-to-End SOC Operations:** Interactive containment and eradication workflows modeled after NIST SP 800-61.

2. Simulation Telemetry & Behavioral Metrics

Metric	Count / Rate	Assessment
Total Emails Simulated	10 Messages	Full synthetic cohort
Emails Opened	8 (80.0%)	Standard engagement level
Simulated Links Clicked	4 (40.0%)	High vulnerability in Finance/Ops
Phishing Reports Submitted	6 (60.0%)	Strong human defense sensor
Awareness Training Completed	4 (100.0% of clickers)	Remediation completed
Incident Time-to-Detect (TTD)	2m 18s	Threshold anomaly rule triggered

Incident Time-to-Contain (TTC)	2m 15s	Campaign disabled & sender blocked
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3. Incident Response Playbook (Incident INC-001)

Simulated Incident INC-001 demonstrated the complete 6-stage lifecycle following NIST SP 800-61:

- 1. Detection:** Rule PHISH-SIM-RULE-104 detected interaction anomalies (≥3 clicks) and opened ticket INC-001.
- 2. Analysis:** Correlated 4 click events and 6 user reports to isolate the impersonated sender profile.
- 3. Containment:** Campaign status flipped to CONTAINED, mail sender blocked, and participant advisory broadcasted.
- 4. Eradication:** Phishing verification URL quarantined and simulation telemetry hashes locked.
- 5. Recovery:** Baseline environment diagnostics verified clean and remedial awareness training assigned.
- 6. Lessons Learned:** Debrief documented human vulnerabilities to artificial urgency and external display names.

4. Strategic Mitigations & Recommendations

- **Hardware-Backed MFA:** Enforce FIDO2 / WebAuthn tokens to neutralize credential theft attempts.
- **Email Authentication:** Enforce strict DMARC (p=reject), DKIM, and SPF protocols.
- **External Email Banners:** Automatically tag inbound external mail to defeat display-name spoofing.
- **One-Click Reporting:** Integrate intuitive report buttons directly within corporate email clients.
- **Continuous Simulation Drills:** Schedule randomized monthly phishing exercises across departments.

5. Conclusion & Verification

PhishAware demonstrates that human-centered security and rapid operational incident response must operate in tandem. The project successfully achieved 100% of the internship capstone requirements, providing clean, runnable source code, reproducible demo data, thorough technical documentation, and presentation collateral.

Submitted By: Cybersecurity Intern	Evaluated By: Internship Supervisor / SOC Lead
Signature: <i>[Signed Electronically]</i>	Approval: Capstone Verified & Accepted